CARTESIAN SQUARES AND TWIST COMPATIBILITIES OVER NESTED AFFINES OF A RELATIVE PROJ

THE STACKS PROJECT AUTHORS

Allowed standard prerequisites (editorial).

- Sections of a quasi-coherent algebra. schemes-lemma-widetilde-pullback (cross-chapter, named). R tensor_{R-prime} A-prime to A is an isomorphism of graded rings.
- Base change morphisms on Proj. lemma-base-change-map-proj (same chapter, not carded). Supplies the morphism r on Proj and the canonical theta isomorphisms of the twists compatible with multiplication.

Source reference. Original lemma 01N2: sources/stacks-constructions/pinned-original-source.tex, source lines 2462–2462. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01I9: focused-seeds/geometry/S9-cons-proj-inclusion/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01NN: sources/stacks-constructions/pinned-original-source.tex, source lines 3580–3580. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *In Situation 01NN. Suppose $U \subset U' \subset S$ are affine opens. Let $A = \mathcal{A}(U)$ and $A' = \mathcal{A}(U')$. The map of graded rings $A' \to A$ induces a morphism $r : Proj(A) \to Proj(A')$, and the diagram*

$$\begin{array}{ccc} Proj(A) & \longrightarrow & Proj(A') \\ \downarrow & & \downarrow \\ U & \longrightarrow & U' \end{array}$$

is cartesian. Moreover there are canonical isomorphisms $\theta : r^*\mathcal{O}_{Proj(A')}(n) \to \mathcal{O}_{Proj(A)}(n)$ compatible with multiplication maps.

Proof. Let $R = \mathcal{O}_S(U)$ and $R' = \mathcal{O}_S(U')$. Note that the map $R \otimes_{R'} A' \to A$ is an isomorphism as $\mathcal{A}$ is quasi-coherent (see Schemes, Lemma 01I9 for example). Hence the lemma follows from Lemma 01N2. □

Date: stacks-project:constructions.tex:local-snapshot-20261012; GNU FDL 1.2 or later.